Team Solutions

1. $2010 = 2 \times 3 \times 5 \times 67$, so 2010 has 16 factors. Eight of them are going to be multiples of 67, eight are not. The smallest of the former is 67, the largest of the latter is 30. Their average is $\boxed{97/2}$ or 48.5.

2. In any rectangle, the sum of the squares of the distances from an interior point to opposite vertices is invariant, i.e., this sum is the same for both pairs of opposite vertices. Thus, $PA^2 + PC^2 = PB^2 + PD^2$ and we have $56^2 + 33^2 = 25^2 + PD^2 \rightarrow 3136 + 1089 = 625 + PD^2 \rightarrow PD^2 = 3600 \rightarrow PD = \boxed{60}$.

   Alternate Solution: Suppose the point P were located on the side $\overline{AB}$. Then $\triangle PBC$ is a right triangle and, using the Pythagorean Theorem, $BC^2 = 464$. Now $\triangle PAD$ is also a right triangle. Therefore, $PD^2 = 56^2 + 464 = 3600 \rightarrow PD = 60$. This solution assumes that the solution is invariant for points P located in the interior of the rectangle OR on the rectangle. A nice topic for discussion!

3. The probability distribution on number of fingers per hand is $(0.1, 0.2, 0.4, 0.2, 0.1)$. By symmetry, the probability of having $10 + k$ fingers is equal to the probability of having $10 - k$ fingers. The probability of having 10 fingers is $0.1^2 + 0.2^2 + 0.4^2 + 0.2^2 + 0.1^2 = 0.26$. The probability of having greater than 10 fingers is $(1 - 0.26)/2 = 0.37$, so the probability of having at least 10 fingers is $0.26 + 0.37 = \boxed{0.63}$.

   Note: We apologize to any nuclear powerplant workers offended by this question. Please don't come after us. After all, we can see you glowing from a mile away.

4. Let $T(n)$ denote the number of ways a $1 \times n$ rectangle can be tiled using these tiles. $T(n)$ satisfies the recursion relation $T(n) = T(n-1) + T(n-2) + T(n-3)$ for all positive n with the initial condition $T(0) = 1$, $T(n) = 0$ for $n < 0$ (consider the size of the rightmost tile of any tiling). The first few values of $T(n)$ for non-negative n are 1, 1, 2, 4, 7, 13, 24, 44, $\boxed{81}$.

5. We want the values of x and y to be alternately as close and as far apart as possible, i.e., maximize and minimize $|x - y|$. Clearly, the maximum occurs when $x = 240$ and $y = 1$ (or vice versa).

   Factoring, $240 = 2^4 \times 3 \times 5$ and $x = 16$ and $y = 15$.

   $x^2 - y^2 = (x+y)(x-y) \rightarrow \begin{cases} x + y = 240 \\ x - y = 1 \end{cases}$ or $\begin{cases} x + y = 16 \\ x - y = 15 \end{cases}$, but each of these systems produce non-integer values of x and y. Clearly, for integer values of x and y, the parity of $x+y$ and $x-y$ must be the same. Revising our choices for x and y $\begin{cases} x + y = 120 \\ x - y = 2 \end{cases}$ or $\begin{cases} x + y = 20 \\ x - y = 12 \end{cases} \rightarrow$ $(x, y) = (61, 59)$ or $(16, 4)$.

   $61(59) - 16(4) = (60 + 1)(60 - 1) - 64 = 3600 - 1 - 64 = \boxed{3535}$.

6. Since angles are preserved upon hitting walls, we can use a series of reflections to express the ball's path as a straight line through a triangular tiling of the plane. In Figure 1 below,

$\triangle ABC$ is at the top and the triangles below contain the number of walls the ball must pass through (bounce off of, in the original triangle) in order to enter:

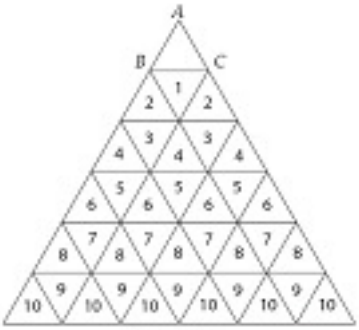


Figure 1

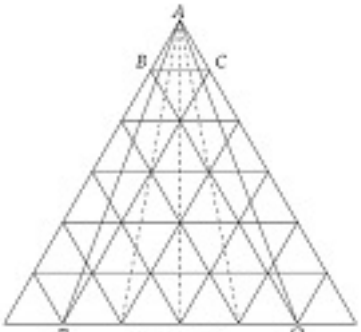
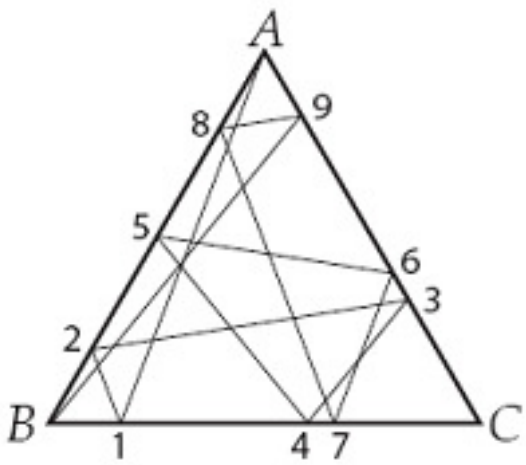


Figure 2

Of the vertices in the '9-bounce' row of triangles, all but two of them are impossible to reach (see the dotted lines in Figure 2), since they correspond to paths that pass through at least one corner of $\triangle ABC$ previously.

The two vertices that are reachable, P and Q, are both equidistant from A (the solid lines in Figure 2); hence the total distance travelled is $AP = AQ = \sqrt{\left(3\sqrt{3}\right)^2 + 2^2} = \boxed{\sqrt{31}}$.

A picture of the ball's path (in the direction towards P) is shown below. The path towards Q is its mirror image:

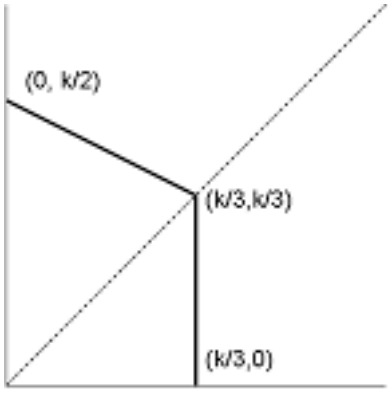


7. The volume of a pyramid is $\frac{Bh}{3}$, where B is the area of the base and h is the height. The hexagon has area $6 \times \frac{\sqrt{3}}{4}$, so the volume of the pyramid is $\frac{\sqrt{3}}{2}$. We can also dissect the pyramid into 7 smaller pyramids whose bases are the 7 faces of the original pyramid, and whose apexes (apices?) are the center of the inscribed sphere. Since the height of each of these pyramids is r, the radius of the sphere, we have $\frac{\sqrt{3}}{2} = \frac{1}{3} \times r \times \left(6 \times \frac{\sqrt{3}}{4} + 6L\right)$, where L is the area of one of the 6 lateral faces of the pyramid. Each of these faces is an isosceles triangle with

base 1 and leg $\sqrt{2}$. We then have a height of $\frac{\sqrt{7}}{2}$, and so $L = \frac{\sqrt{7}}{4}$. Solving for r we get $\frac{r}{3}\left(6 \cdot \frac{\sqrt{3}}{4} + 6 \cdot \frac{\sqrt{7}}{4}\right) = \frac{\sqrt{3}}{2} \to r = \frac{\sqrt{3}}{\sqrt{3}+\sqrt{7}} = \boxed{\frac{\sqrt{21}-3}{4}}$.

8. $f(x) = \frac{x^4+3x^3-4x^2}{x^2+4x-4} = \frac{x^2(x+4)(x-1)}{\left(x+(2-2\sqrt{2})\right)\left(x+(2+2\sqrt{2})\right)}$. $f(x)$ has zeroes at 0, 1, and -4 and asymptotes at $-2-2\sqrt{2}$ and $-2+2\sqrt{2}$ (approximately -4.8 and 0.8). Testing the sign of a single point in the interval between two zeroes or asymptotes give you the sign of the function over the entire interval. The function is negative between $-2-2\sqrt{2}$ and -4 as well as between $-2+2\sqrt{2}$ and 1. The sum of the lengths of these intervals is $\left(1-\left(2\sqrt{2}-2\right)\right) + \left(-4-\left(-2\sqrt{2}-2\right)\right) = 1-2\sqrt{2}+2-4+2\sqrt{2}+2 = \boxed{1}$.

9. Noting that the function $f(x,y) = |x|+|x-y|+|x+y|$ is symmetric about both the x- and y-axis, it suffices to compute value of k for which the area in the first quadrant is 5/2. When $x \geq y$, $|x|+|x-y|+|x+y| = x+x-y+x+y = 3x$. When $x \leq y$, $|x|+|x-y|+|x+y| = x+y-x+x+y = x+2y$. For a fixed value of k, this function looks like the graphic below in the first quadrant. The area of the region is $\frac{1}{2}\left(\frac{k}{2}+\frac{k}{3}\right)\frac{k}{3} = \frac{5k^2}{36} = \frac{5}{2} \to k = \boxed{3\sqrt{2}}$.



10. $E \times C$ and $D \times C$ both end in C, and the only non-zero C for which that is possible for two different digits is 5. E and D must be odd, so by looking in the thousands column, we conclude that D is 3. From that same column we conclude that $D \times A = D$, so A must be 1. $1\underline{B}5 \times 3 = 3\underline{E}5$, and the only distinct non-zero B and odd E that hold are $B = 2$ and $E = 7$. $125 \times 7 = \underline{F}75$, so $F = 8$. Therefore, $\underline{A}\,\underline{B}\,\underline{C}\,\underline{D}\,\underline{E}\,\underline{F} = \boxed{125378}$. The completed multiplication is given below.

$$\begin{array}{cccc} & 1 & 2 & 5 \\ \times & 3 & 7 & \\ & 8 & 7 & 5 \\ 3 & 7 & 5 & \\ \hline 4 & 6 & 2 & 5 \end{array}$$